Homework 1. Who to sit down with this week

70-467 - Machine Learning for Business Analytics

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2026-09-08

Background. A clothing factory in Bangladesh records, each day, how productive each team was. Sewing is stitching. Finishing is later: trim, press, pack. Each team has a posted target (`targeted_productivity`) and a recorded rate (`actual_productivity`). Both are rates, not a count of shirts. 1.0 would mean they delivered what the standard time for that style said they should. In this file the target never exceeds 0.80. Actual productivity sometimes goes above 1.

The sheet also has overtime (minutes for the team that day), a cash incentive in Bangladeshi taka, SMV (minutes the operation is supposed to take), and WIP (unfinished pieces on the sewing line; missing for finishing). The codebook on the next page gives type and unit for every column. There is no Friday. Department comes out as `sweing`, `finishing`, and `finishing ` with a space. `read.csv` shows all three; `readr` by default hides the space.

The problem. You have 1,197 team-days, 1 January to 11 March 2015. Some teams miss the target. A lot of finishing teams do. Someone has to sit down with teams this week and decide what to talk about: overtime, the incentive cheque, the target, or something the sheet does not record.

Please say who (finishing versus sewing, and which teams), and whether overtime or the incentive cheque still belongs in that conversation. Please use the starter and modify it accordingly.

Deliverables.

This homework is long. Please start early. Submit on Canvas, Tuesday 15 September 2026, 11:59 pm:

- a runnable R notebook,
- at most six slides, and
- a video of at most five minutes.

After you lock a model, the starter also asks you to run additional diagnosis to determine the validity of what you are doing. Please make sure you also complete that.

About the video.

- Please record a video of at most five minutes, with at most six slides.
- The camera must be on. I need to see you delivering.
- Start with what the plant is trying to decide. Then how you approached it. Then one or two results. Then who you would sit down with this week, and whether overtime or the cheque is worth that meeting. Please also include any other concluding remark you deem relevant to the conversation.

The starter has what to submit and the common mistakes.

Important. Please make sure the sound and video are good when preparing your delivery. Please don't read slides or notes; this must be your natural presentation.

If you read the slides, that is a zero on the slides and the video. AI may help you prepare; it must not deliver the video. Do not say that raising overtime or the cheque will lift actual productivity.

Codebook. 1,197 rows, 15 columns. One row = one team on one calendar day.

Variable	Type	Unit	Description
date	date	calendar day	Day of the record (m/d/yyyy). 1 Jan to 11 Mar 2015.
quarter	factor	none	Not a fiscal quarter. A month split into four blocks. Quarter5 is 44 leftover rows, not a fifth season.
department	text	none	Sewing stitches; finishing is later (trim, press, pack). Raw: `sweing`, `finishing`, and `finishing ` with a space. `read.csv` shows all three; `readr` default hides the space.
day	factor	none	Weekday. No Friday in this file.
team	numeric	team id	Team 1 to 12. The same number can appear in sewing and in finishing.
targeted_productivity	numeric	rate	Target posted for that team that day. Never exceeds 0.80 here. Not the outcome.
smv	numeric	minutes	Standard minute value: minutes the style is supposed to take. About 3 to 55 minutes here.
wip	numeric	pieces	Unfinished garments still on the sewing line. Filled for sewing; missing on every finishing row. A pooled `lm` with `wip` drops finishing.
over_time	numeric	minutes	Extra minutes for the team that day, not one person. Median 3,960 minutes.
incentive	numeric	BDT	Cash in Bangladeshi taka. Often 0; often about 50 when not. Ten finishing days on 9 March are 960 to 3,600.
idle_time	numeric	minutes	Minutes the line was stopped. Positive on 18 sewing rows only.
idle_men	numeric	workers	Workers idle during that stoppage. Same 18 rows.
no_of_style_change	numeric	count	Times the team switched to a different style that day. 0 on most rows.
no_of_workers	numeric	count	People on the team that day. About 2 to 89 (median 34).
actual_productivity	numeric	rate	What they delivered, as a fraction of the standard. This is Y. Mean about 0.74. Some rows exceed 1.